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UNet++: A Nested U-Net Architecture for Medical Image Segmentation

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U-Net++

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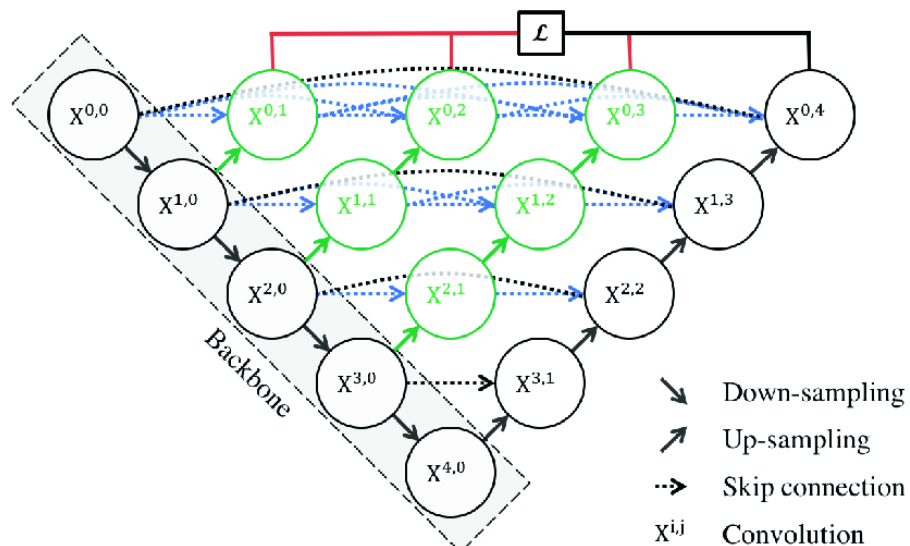
Introduction

Background

- U-Net and FCN performs well in segmentation and they were SOTA.
- Segmenting lesions or abnormalities in medical images demands a higher level of accuracy than what is desired in natural images.
- U-Net and FCN don't meet this.
- To address this problem U-Net++ introduces **nested** and **dense** skip connection.

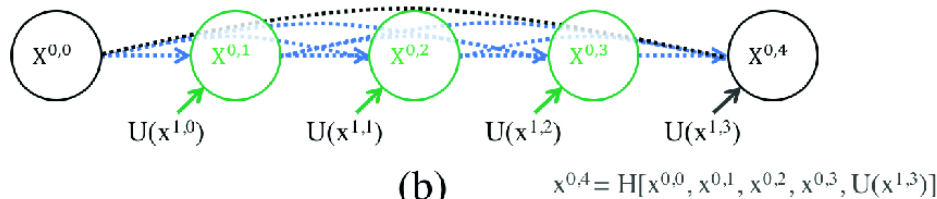
Method

Proposed Network Architecture: UNet++

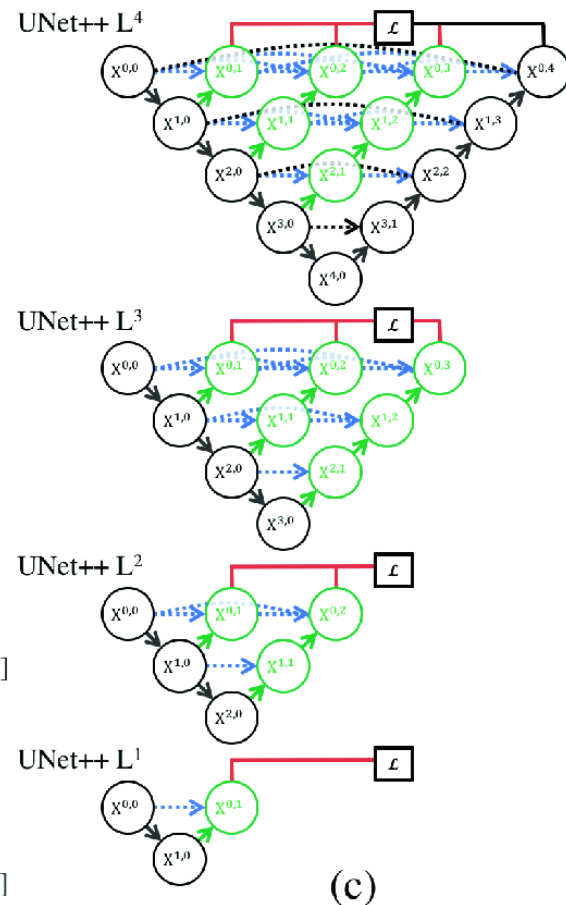


(a)

$$x^{0,1} = H[x^{0,0}, U(x^{1,0})] \quad x^{0,2} = H[x^{0,0}, x^{0,1}, U(x^{1,1})] \quad x^{0,3} = H[x^{0,0}, x^{0,1}, x^{0,2}, U(x^{1,2})]$$



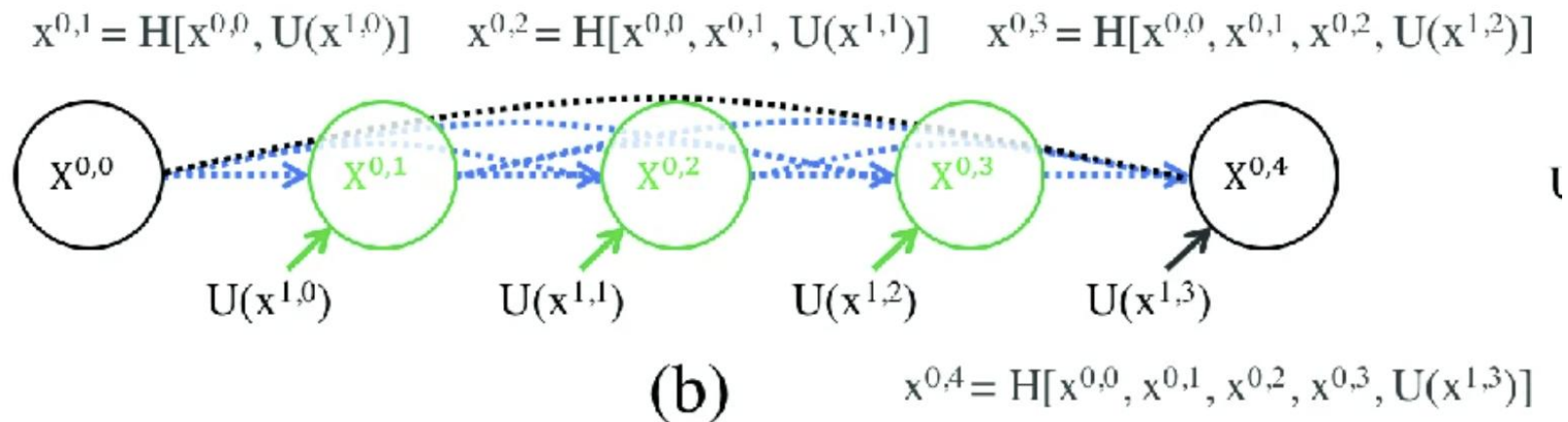
(b)



(c)

Method

Re-designed Skip Pathways



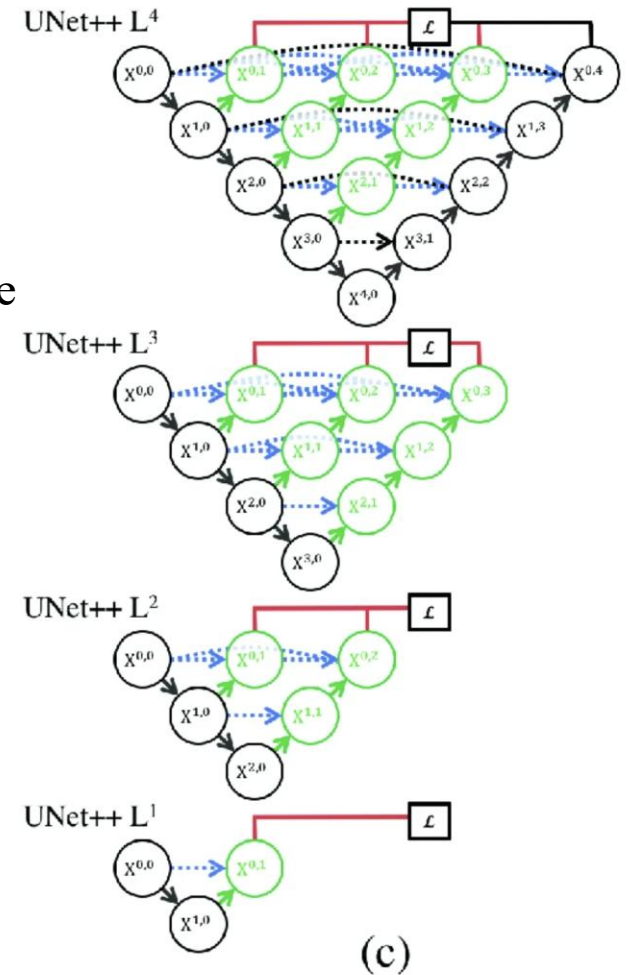
$$x^{i,j} = \begin{cases} \mathcal{H}(x^{i-1,j}), & j = 0 \\ \mathcal{H}\left(\left[[x^{i,k}]_{k=0}^{j-1}, \mathcal{U}(x^{i+1,j-1})\right]\right), & j > 0 \end{cases}$$

Method

Deep Supervision

- UNet++ differs from U-Net in three ways:
- Convolution layers on skip pathways reduce the semantic gap between encoder and decoder features.
- Dense skip connections on skip pathways improve gradient flow.
- Deep supervision enables model pruning and maintains or improves performance compared to using a single loss layer.

$$\mathcal{L}(Y, \hat{Y}) = -\frac{1}{N} \sum_{b=1}^N \left(\frac{1}{2} \cdot Y_b \cdot \log \hat{Y}_b + \frac{2 \cdot Y_b \cdot \hat{Y}_b}{Y_b + \hat{Y}_b} \right)$$



Experiments

Experiment detail

- **Datasets**
 - 4 medical imaging datasets
 - Different modalities and lesions/organs
- **Baselines**
 - Original U-Net & wide U-Net (similar params)
 - To confirm improvement is not due to model size
- **Implementation**
 - Metrics: Dice, IoU
 - Adam optimizer (lr=3e-4), early stopping
 - Deep supervision: 4 outputs averaged

Table 1: The image segmentation datasets used in our experiments.

Dataset	Images	Input Size	Modality	Provider
cell nuclei	670	96×96	microscopy	Data Science Bowl 2018
colon polyp	7,379	224×224	RGB video	ASU-Mayo [10,11]
liver	331	512×512	CT	MICCAI 2018 LiTS Challenge
lung nodule	1,012	64×64×64	CT	LIDC-IDRI [1]

Experiments

result

■ Results

- Wide U-Net > U-Net (except liver)
- UNet++ (no deep supervision) +2.8~3.3 IoU gain vs. others
- UNet++ (with deep supervision) +0.6 IoU gain vs. without
- Deep supervision helps liver & lung; less effect on nuclei & polyp

Table 3: Segmentation results (IoU: %) for U-Net, wide U-Net and our suggested architecture UNet++ with and without deep supervision (DS).

Architecture	Params	Dataset			
		cell nuclei	colon polyp	liver	lung nodule
U-Net [9]	7.76M	90.77	30.08	76.62	71.47
Wide U-Net	9.13M	90.92	30.14	76.58	73.38
UNet++ w/o DS	9.04M	92.63	33.45	79.70	76.44
UNet++ w/ DS	9.04M	92.52	32.12	82.90	77.21

Experiments

result

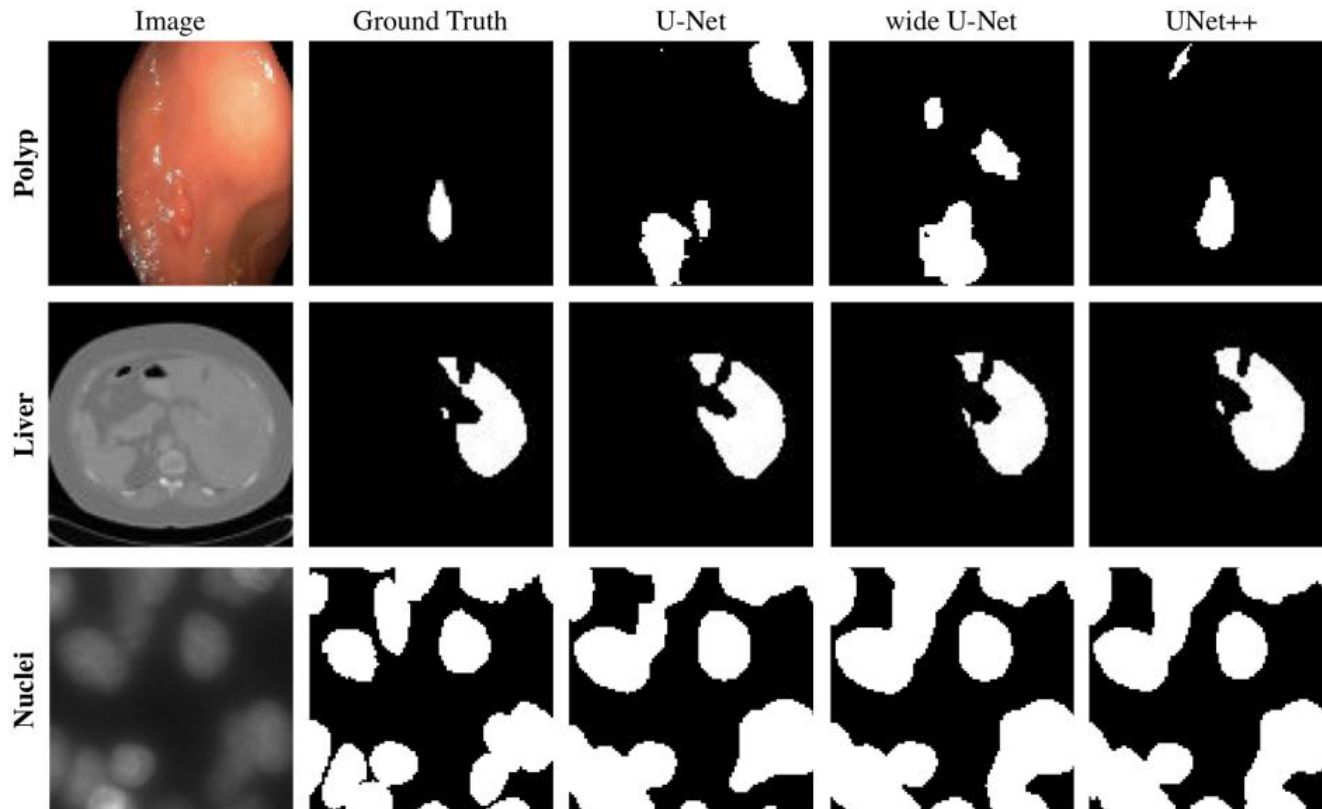


Fig. 2: Qualitative comparison between U-Net, wide U-Net, and UNet++, showing segmentation results for polyp, liver, and cell nuclei datasets (2D-only for a distinct visualization).

Experiments

Model Pruning

■ Model Pruning

- UNet++ pruning reduces inference time by ~32%
- IoU drops only 0.6 points on average
- More pruning → faster but less accurate

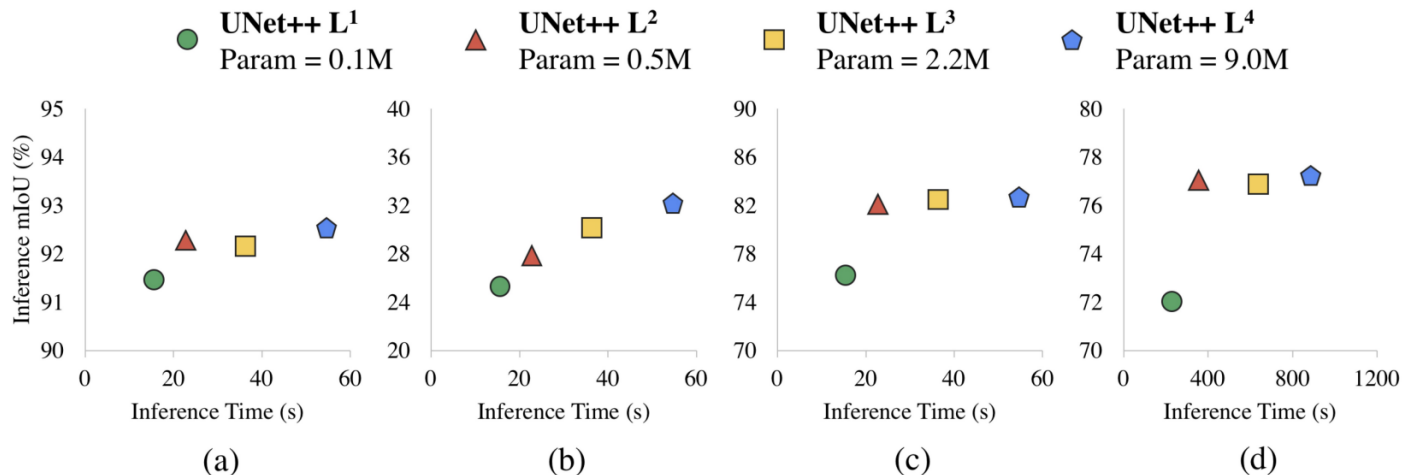


Fig. 3: Complexity, speed, and accuracy of UNet++ after pruning on (a) cell nuclei, (b) colon polyp, (c) liver, and (d) lung nodule segmentation tasks respectively. The inference time is the time taken to process **10k** test images using one NVIDIA TITAN X (Pascal) with 12 GB memory.

Conclusion

UNet++ for Accurate Medical Image Segmentation

- Uses redesigned skip pathways to reduce semantic gap between encoder and decoder feature maps, simplifying optimization.
- Applies deep supervision to improve segmentation accuracy, especially for multi-scale lesions like colon polyps.
- Evaluated on four datasets: lung nodules, colon polyps, cell nuclei, and liver segmentation.
- Achieved average IoU improvements of 3.9 and 3.4 points over U-Net and wide U-Net, respectively.